

Project 2

EXPLORATORY DATA ANALYSIS (EDA)

DecodeLabs Data Analytics Internship

Prepared By

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Tools Used

Microsoft Excel

Pivot Tables

Power Query

Microsoft Word

GitHub

Project Objective

To perform Exploratory Data Analysis (EDA) on the provided dataset in order to identify patterns, trends, distributions, and outliers using descriptive statistics and analytical techniques.

Submission Date:

24 May 2026

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Project Overview

This project was completed as part of the DecodeLabs Data Analytics Internship Project 2. The objective of this project is to perform Exploratory Data Analysis (EDA) on a raw dataset to identify meaningful patterns, trends, distributions, and outliers.

The project focuses on descriptive statistics, data quality verification, trend analysis, and business insights using Microsoft Excel and Power Query.

Dataset Overview

The dataset contains e-commerce order information including product details, customer information, payment methods, order status, coupon usage, referral sources, and pricing data.

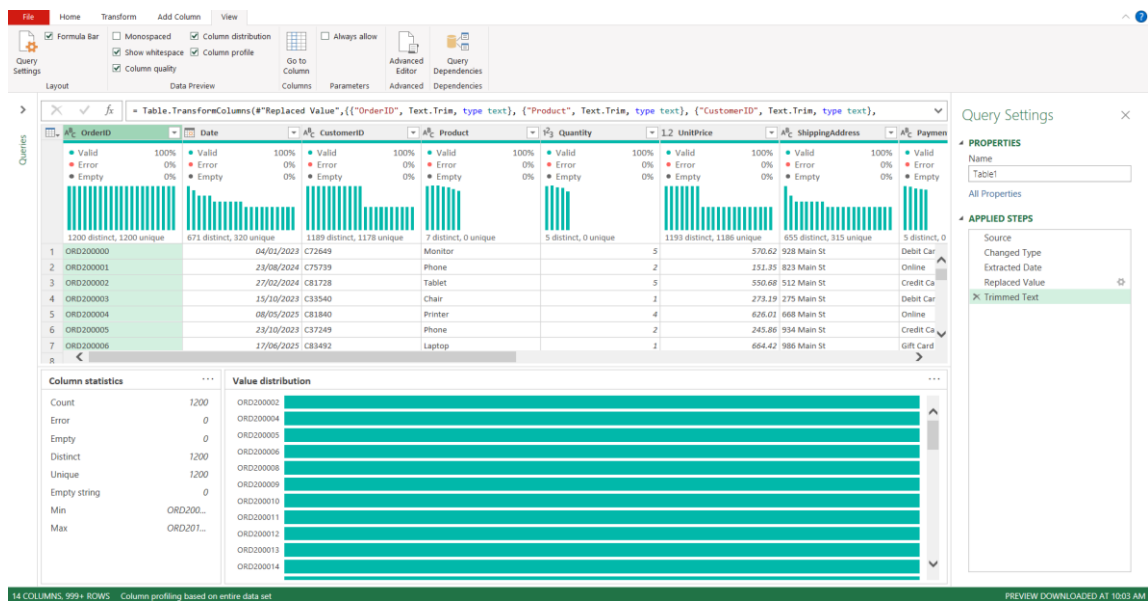
The dataset was analyzed to understand customer purchasing behavior, payment preferences, order outcomes, and overall business trends.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
2	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947303	7	SAVE10	Instagram	2853.1
3	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
4	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903082	8	FREESHIP	Email	2753.4
5	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
6	ORD200004	2023-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
7	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
8	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
9	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
10	ORD200008	2025-04-02	C36817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2		Email	266.56
11	ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
12	ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
13	ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
14	ORD200012	2024-10-15	C38785	Monitor	2	180.55	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
15	ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
16	ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
17	ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54920938	3		Google	473.96
18	ORD200016	2023-01-24	C28976	Printer	2	533.61	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
19	ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3		Referral	423.4
20	ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
21	ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120
22	ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
23	ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6		Google	1371.8
24	ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
25	ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093
26	ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.9
27	ORD200025	2024-04-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
28	ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6		Referral	297.75
29	ORD200027	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74
30	ORD200028	2024-11-19	C60873	Phone	4	160.07	848 Main St	Online	Shipped	TRK65808530	6		Email	640.28

Data Quality Verification

Before starting the exploratory analysis, the dataset was inspected to verify data quality. Data profiling tools in Power Query were used to identify missing values, duplicate records, incorrect data types, and formatting inconsistencies.

The inspection confirmed that the dataset required minor preprocessing before analytical activities could begin.



Data Preparation

Before performing exploratory data analysis, the dataset was prepared and reviewed to ensure consistency and analytical accuracy.

The preparation process included validation of records, verification of data structure, handling of missing values where required, and standardization of data formats. After preparation, a cleaned dataset was generated and used for all subsequent analyses.

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200003	15/10/2023	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200020	08/05/2023	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
ORD200031	30/06/2023	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6	No Coupon	Facebook	1500.24
ORD200033	05/06/2023	C52184	Chair	1	162.77	912 Main St	Online	Cancelled	TRK73312471	4	WINTER15	Facebook	162.77
ORD200034	07/01/2023	C67195	Chair	4	576.52	926 Main St	Credit Card	Pending	TRK55467915	5	WINTER15	Facebook	2306.08
ORD200043	14/12/2023	C60572	Chair	3	499.21	871 Main St	Online	Cancelled	TRK20351666	8	No Coupon	Instagram	1497.63
ORD200045	10/02/2024	C17388	Chair	1	317.25	157 Main St	Debit Card	Returned	TRK47077395	1	SAVE10	Facebook	317.25
ORD200054	27/03/2025	C48120	Chair	1	280.25	623 Main St	Online	Pending	TRK11014449	5	WINTER15	Email	280.25
ORD200059	11/09/2024	C97603	Chair	4	593.27	811 Main St	Online	Shipped	TRK47055028	5	No Coupon	Facebook	2373.08
ORD200063	02/02/2025	C96695	Chair	1	564.86	813 Main St	Debit Card	Shipped	TRK64283851	6	WINTER15	Referral	564.86
ORD200066	15/11/2024	C18104	Chair	5	127.18	715 Main St	Cash	Cancelled	TRK15488568	9	SAVE10	Referral	635.9
ORD200068	11/12/2023	C41598	Chair	3	490.92	709 Main St	Credit Card	Pending	TRK81865940	5	FREESHIP	Email	1472.76
ORD200090	18/09/2023	C36540	Chair	4	182.08	611 Main St	Gift Card	Delivered	TRK35828122	4	SAVE10	Email	728.32
ORD200094	11/02/2023	C79821	Chair	1	531.18	781 Main St	Gift Card	Returned	TRK12780193	6	WINTER15	Instagram	531.18
ORD200133	20/06/2025	C87432	Chair	4	302.05	636 Main St	Debit Card	Cancelled	TRK68969625	6	SAVE10	Google	1208.2
ORD200140	31/05/2025	C33110	Chair	3	528.14	388 Main St	Online	Delivered	TRK38624225	3	FREESHIP	Email	1584.42
ORD200143	13/06/2024	C66521	Chair	4	400.07	933 Main St	Online	Shipped	TRK34654244	5	No Coupon	Referral	1600.28
ORD200152	02/01/2025	C77736	Chair	2	399.04	659 Main St	Cash	Cancelled	TRK90329250	3	FREESHIP	Referral	798.08
ORD200153	18/12/2024	C71570	Chair	4	534.48	437 Main St	Online	Returned	TRK95992269	9	WINTER15	Instagram	2137.92
ORD200156	08/11/2024	C58808	Chair	2	269.83	651 Main St	Online	Delivered	TRK24451893	4	SAVE10	Email	539.66
ORD200162	27/07/2023	C82313	Chair	3	65.86	917 Main St	Online	Pending	TRK79854624	8	WINTER15	Google	197.58
ORD200166	04/03/2025	C28889	Chair	3	86.29	565 Main St	Cash	Delivered	TRK23791726	6	WINTER15	Instagram	258.87
ORD200167	27/05/2023	C88304	Chair	2	384.14	219 Main St	Online	Returned	TRK44268941	2	WINTER15	Email	768.28
ORD200175	08/02/2025	C78170	Chair	5	68.34	332 Main St	Debit Card	Pending	TRK71410605	6	WINTER15	Referral	341.7
ORD200177	05/08/2023	C55632	Chair	5	431.69	354 Main St	Credit Card	Pending	TRK13003066	10	No Coupon	Facebook	2158.45
ORD200178	11/04/2024	C53070	Chair	4	512.06	283 Main St	Online	Delivered	TRK49791755	5	SAVE10	Google	2048.24
ORD200187	08/02/2023	C21227	Chair	1	590.87	689 Main St	Gift Card	Returned	TRK79129657	6	FREESHIP	Referral	590.87

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to summarize the main characteristics of the dataset and identify meaningful patterns and trends. The analysis included descriptive statistics, trend analysis, revenue analysis, and outlier detection. Pivot Tables, Pivot Charts, and Excel statistical functions were used to support the analytical process.

Descriptive Statistics Analysis

Descriptive statistics were calculated to summarize the key characteristics of the dataset and provide an overall understanding of the numerical variables.

The analysis revealed that the dataset contains **1,200 records, 7 unique products, 5 payment methods, 5 order statuses, and 5 referral sources.**

The average Unit Price was **356.41**, while the average Total Price was **1053.97**.

These values indicate the general spending pattern of customers within the dataset.

The numerical summary shows that:

- Quantity ranges from **1 to 5 units**
- Unit Price ranges from **11.39 to 699.93**
- Items In Cart ranges from **1 to 10**
- Total Price ranges from **11.39 to 3456.40**

The comparison between mean and median values indicates that most variables have relatively stable distributions, while Total Price shows greater variation due to a small number of high-value transactions.

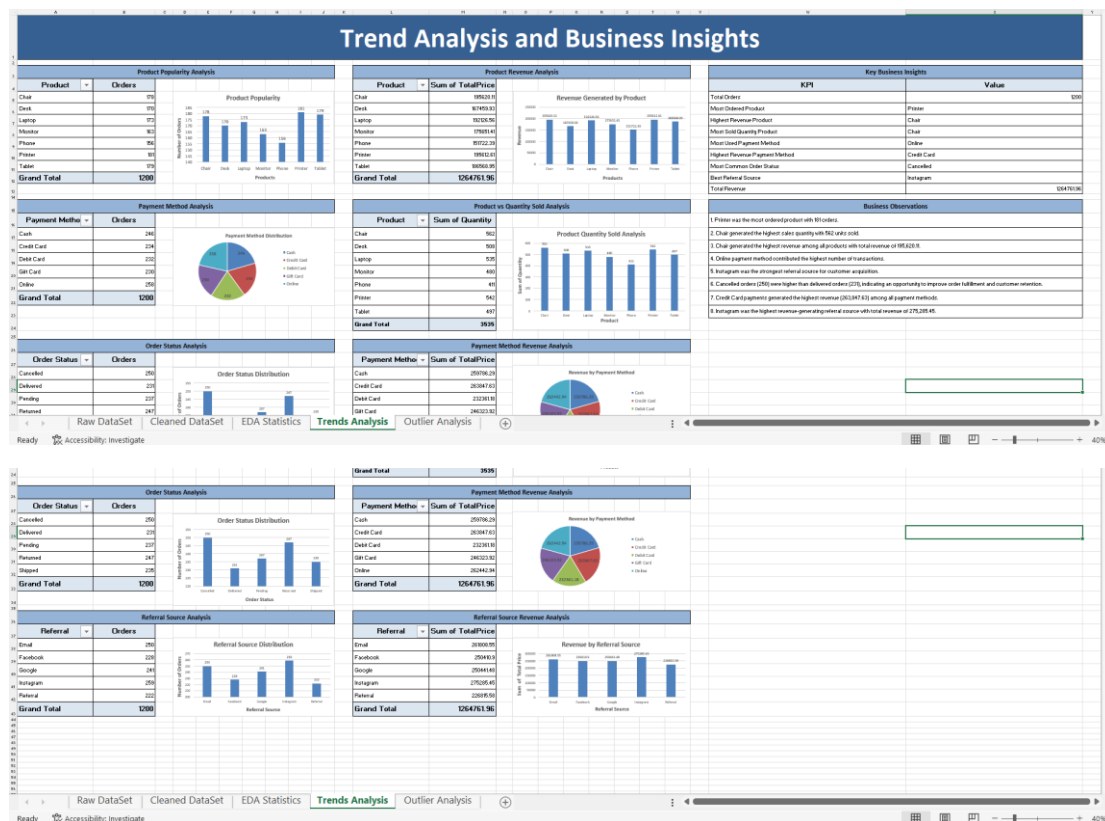
Exploratory Data Analysis (EDA) - Descriptive Statistics					
Metric	Value				
Total Records	1200				
Total Products	7				
Total Payment Methods	5				
Total Order Statuses	5				
Total Referral Sources	5				
Average Unit Price	356.41				
Average Total Price	1053.97				

Numerical Data Summary					
Metrics	Min	Max	Mean	Median	
Quantity	1	5	2.95	3	
Unit Price	11.39	699.93	356.41	364.21	
Item In Cart	1	10	5.49	5	
Total Price	11.39	3456.40	1053.97	823.62	

Trend Analysis

Trend analysis was performed to identify patterns in customer purchasing behavior, product performance, revenue generation, payment preferences, referral effectiveness, and order outcomes.

Pivot Tables and Pivot Charts were used to summarize and visualize business trends within the dataset. The analysis helped identify high-performing products, preferred payment methods, major referral sources, and revenue-driving factors.

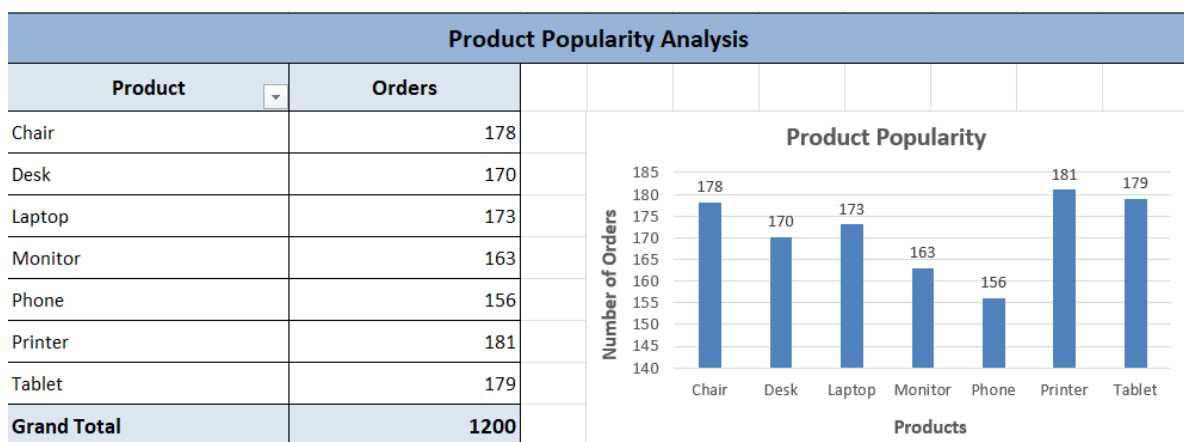


Product Popularity Analysis

Product popularity was evaluated based on the number of orders received by each product.

The analysis revealed that **Printer** was the most frequently ordered product with **181 orders**, followed by **Tablet (179 orders)** and **Chair (178 orders)**. **Phone** received the lowest number of orders with **156 orders**.

This indicates that printers and tablets were the most preferred products among customers.

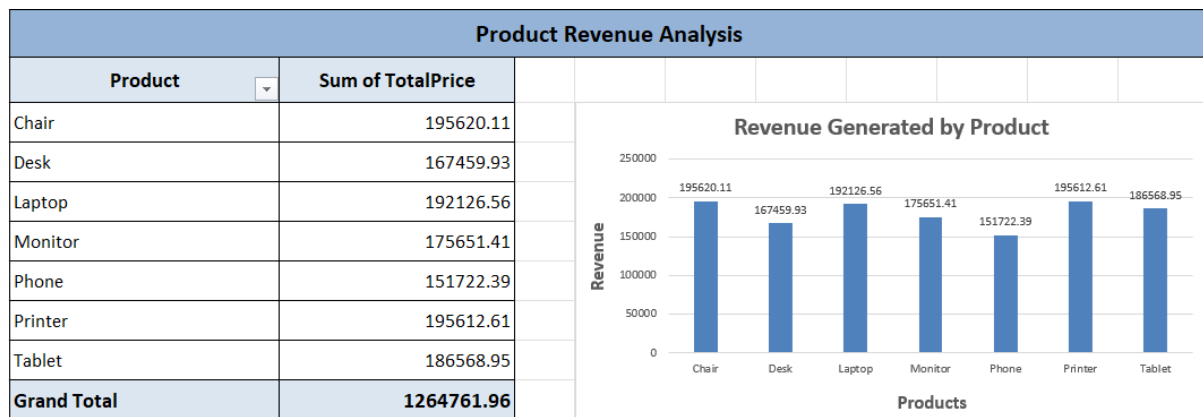


Product Revenue Analysis

Revenue analysis was conducted to determine which products contributed the most to total sales revenue.

The results showed that **Chair generated the highest revenue of 195,620.11**, closely followed by **Printer (195,612.61)** and **Laptop (192,126.56)**.

These products contributed significantly to overall business revenue and represent the strongest-performing product categories.

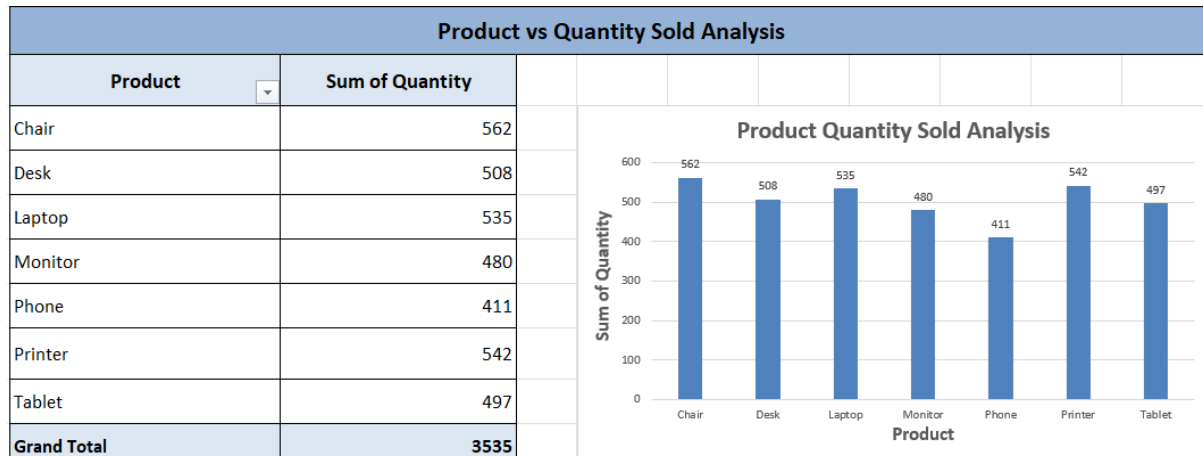


Product Quantity Sold Analysis

Product quantity analysis measured the total number of units sold for each product.

The analysis showed that **Chair recorded the highest sales quantity with 562 units sold**, followed by **Printer (542 units)** and **Laptop (535 units)**.

This suggests that chairs experienced strong customer demand throughout the analyzed period.

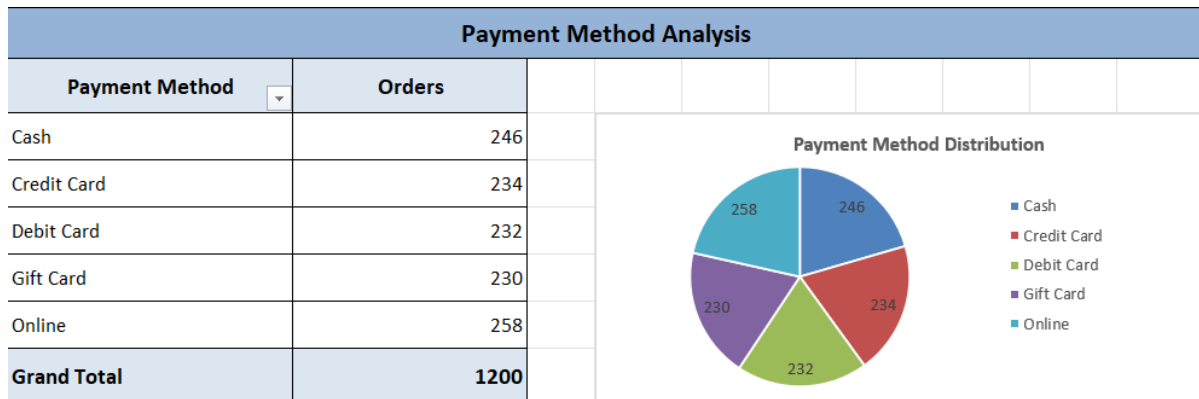


Payment Method Analysis

Payment method analysis was performed to understand customer transaction preferences.

The results indicated that **Online payments were the most frequently used payment method with 258 transactions**, followed by **Cash (246 transactions)**.

This demonstrates a customer preference for digital payment channels.

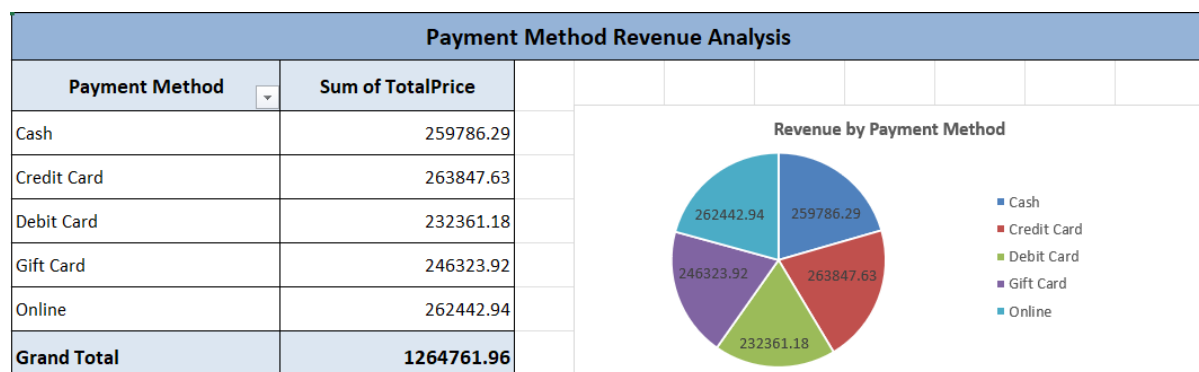


Payment Method Revenue Analysis

Revenue generated by each payment method was analyzed to determine which payment channel contributed the highest sales value.

The analysis revealed that **Credit Card transactions generated the highest revenue of 263,847.63**, followed by **Online payments (262,442.94)**.

This indicates that customers using credit cards generally contributed higher-value purchases.

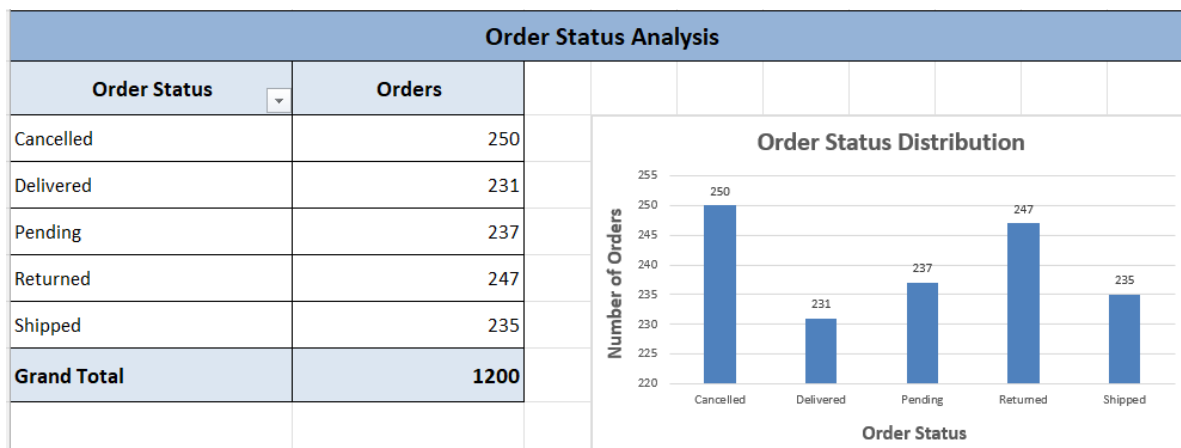


Order Status Analysis

Order status analysis was conducted to understand the distribution of completed and incomplete orders.

The results showed that **Cancelled orders were the most common status with 250 records**, while **Delivered orders accounted for 231 records**.

The difference between cancelled and delivered orders highlights a potential opportunity to improve order fulfillment and customer retention processes.

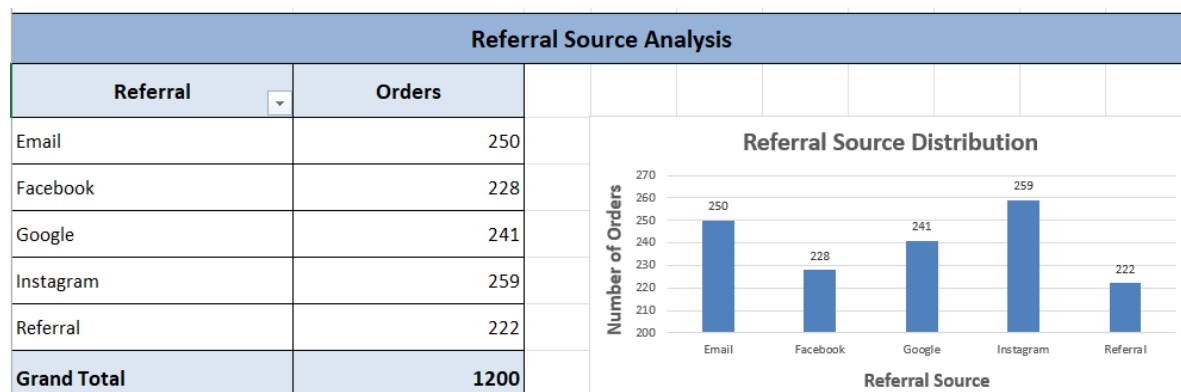


Referral Source Analysis

Referral source analysis examined how customers were acquired through different channels.

The analysis showed that **Instagram generated the highest number of referrals with 259 orders**, followed by **Email (250 orders)** and **Google (241 orders)**.

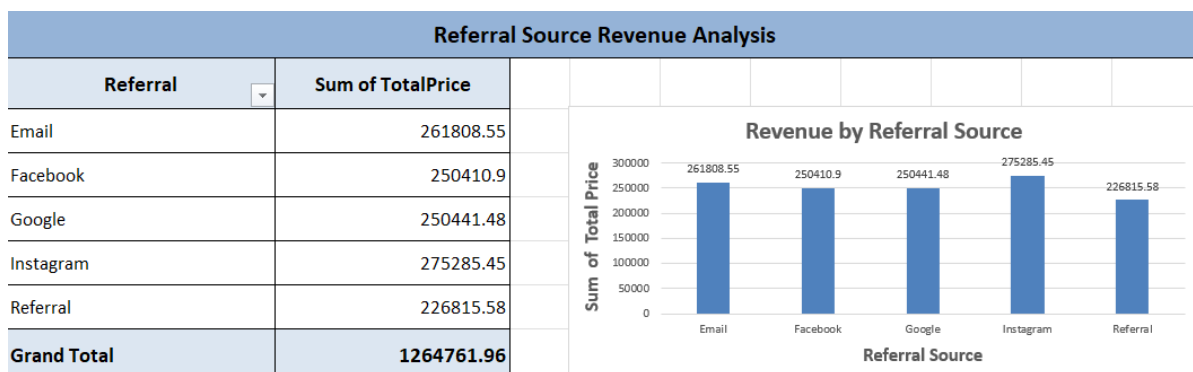
This suggests that Instagram was the most effective customer acquisition channel within the dataset.



Referral Source Revenue Analysis

Revenue generated through referral channels was also evaluated.

The analysis revealed that **Instagram generated the highest referral revenue of 275,285.45**, making it both the most effective referral source and the strongest revenue-generating acquisition channel.



Key Business Insights

The trend analysis identified several important business performance indicators from the dataset.

The analysis revealed that the dataset contains a total of **1,200 orders**. Printer was identified as the most frequently ordered product, while Chair generated the highest revenue and recorded the highest sales quantity.

Online payment was the most commonly used payment method, whereas Credit Card transactions generated the highest overall revenue. Instagram emerged as the strongest referral source for both customer acquisition and revenue generation.

These findings provide valuable insights into customer preferences, product performance, and revenue-driving factors within the business.

Key Business Insights	
KPI	Value
Total Orders	1200
Most Ordered Product	Printer
Highest Revenue Product	Chair
Most Sold Quantity Product	Chair
Most Used Payment Method	Online
Highest Revenue Payment Method	Credit Card
Most Common Order Status	Cancelled
Best Referral Source	Instagram
Total Revenue	1264761.96

Business Observations

Based on the trend analysis, the following observations were identified:

1. Printer was the most ordered product with **181 orders**.
2. Chair generated the highest sales quantity with **562 units sold**.
3. Chair generated the highest revenue among all products with a total revenue of **195,620.11**.
4. Online payment method contributed the highest number of transactions.
5. Instagram was the strongest referral source for customer acquisition.
6. Cancelled orders (**250**) were slightly higher than delivered orders (**231**), indicating an opportunity to improve order fulfillment and customer retention.
7. Credit Card payments generated the highest revenue (**263,847.63**) among all payment methods.
8. Instagram was the highest revenue-generating referral source with total revenue of **275,285.45**.

Business Observations
1. Printer was the most ordered product with 181 orders.
2. Chair generated the highest sales quantity with 562 units sold.
3. Chair generated the highest revenue among all products with total revenue of 195,620.11.
4. Online payment method contributed the highest number of transactions.
5. Instagram was the strongest referral source for customer acquisition.
6. Cancelled orders (250) were higher than delivered orders (231), indicating an opportunity to improve order fulfillment and customer retention.
7. Credit Card payments generated the highest revenue (263,847.63) among all payment methods.
8. Instagram was the highest revenue-generating referral source with total revenue of 275,285.45.

Outlier Analysis

The objective of outlier analysis was to identify unusual or extreme values within the dataset that could potentially influence statistical results and business insights. The Interquartile Range (IQR) method was used to detect outliers in key numerical variables including Quantity, UnitPrice, ItemsInCart, and TotalPrice.

Outlier Analysis					
Quantity Outlier Detection		TotalPrice Outlier Analysis		Outlier Summary	
Metric	Value	Metric	Value	Analysis Type	Result
Q1	2	Q1	410.52	Quantity Outliers	0
Q3	4	Q3	1578.48	UnitPrice Outliers	0
IQR	2	IQR	1167.96	ItemsInCart Outliers	0
Lower Bound	-1	Lower Bound	-1341.41	TotalPrice Outliers	8
Upper Bound	7	Upper Bound	3330.41	Data Suitable for Analysis	Yes
Outliers Count	0	Outliers Count	8	Outliers Retained	Yes
No quantity outliers were detected. All quantity values fall within the acceptable IQR range.		Eight TotalPrice outliers were identified using the IQR method. These represent genuine high-value customer transactions and were retained for business analysis.			
UnitPrice Outlier Analysis		ItemsInCart Outlier Analysis		Business Insights	
Metric	Value	Metric	Value	1. IQR method was used to identify potential outliers.	
Q1	186	Q1	4	2. No outliers were found in Quantity, UnitPrice, or ItemsInCart.	
Q3	522	Q3	7	3. Eight outliers were detected in TotalPrice.	
IQR	336	IQR	3	4. These outliers represent genuine high-value transactions.	
Lower Bound	-317.20	Lower Bound	-0.5	5. The dataset remains suitable for business analysis and reporting.	
Upper Bound	1024.83	Upper Bound	11.5	6. Revenue variation is influenced by a small number of legitimate high-value transactions.	
Outliers Count	0	Outliers Count	0		
No UnitPrice outliers were detected, indicating stable product pricing across transactions.		No ItemsInCart outliers were detected. Customer cart sizes remain within expected business limits.		Series "Total" Point "Laptop" Data Label	

Raw DataSet

Cleaned DataSet

EDA Statistics

Trends Analysis

Outlier Analysis

Methodology

Outliers were identified using the Interquartile Range (IQR) method.

The following calculations were performed:

- Q1 (First Quartile)
- Q3 (Third Quartile)
- $IQR = Q3 - Q1$
- $Lower\ Bound = Q1 - (1.5 \times IQR)$
- $Upper\ Bound = Q3 + (1.5 \times IQR)$

Any value falling below the lower bound or above the upper bound was considered an outlier.

Quantity Outlier Analysis

The Quantity variable was analyzed using the IQR method.

- $Q1 = 2$
- $Q3 = 4$
- $IQR = 2$
- $Lower\ Bound = -1$
- $Upper\ Bound = 7$
- $Outliers\ Detected = 0$

No quantity outliers were detected. All quantity values fall within the acceptable IQR range.

Quantity Outlier Detection	
Metric	Value
Q1	2
Q3	4
IQR	2
Lower Bound	-1
Upper Bound	7
Outliers Count	0
No quantity outliers were detected. All quantity values fall within the acceptable IQR range.	

UnitPrice Outlier Analysis

The UnitPrice variable was evaluated to identify unusual product pricing values.

- Q1 = 186
- Q3 = 522
- IQR = 336
- Lower Bound = -317.20
- Upper Bound = 1024.83
- Outliers Detected = 0

No UnitPrice outliers were identified, indicating that product pricing remains consistent across transactions.

UnitPrice Outlier Analysis	
Metric	Value
Q1	186
Q3	522
IQR	336
Lower Bound	-317.20
Upper Bound	1024.83
Outliers Count	0
No UnitPrice outliers were detected, indicating stable product pricing across transactions.	

ItemsInCart Outlier Analysis

The ItemsInCart variable was analyzed to identify unusual cart sizes.

- Q1 = 4
- Q3 = 7
- IQR = 3
- Lower Bound = -0.5
- Upper Bound = 11.5
- Outliers Detected = 0

No outliers were detected for ItemsInCart, indicating that customer cart sizes remain within expected business limits.

ItemsInCart Outlier Analysis	
Metric	Value
Q1	4
Q3	7
IQR	3
Lower Bound	-0.5
Upper Bound	11.5
Outliers Count	0
No ItemsInCart outliers were detected. Customer cart sizes remain within expected business limits.	

TotalPrice Outlier Analysis

The TotalPrice variable was analyzed to identify unusually high or low transaction values.

- Q1 = 410.52
- Q3 = 1578.48
- IQR = 1167.96
- Lower Bound = -1341.41
- Upper Bound = 3330.41
- Outliers Detected = 8

A total of eight outliers were detected in the TotalPrice column. These records represent legitimate high-value customer purchases and were retained for business analysis.

TotalPrice Outlier Analysis	
Metric	Value
Q1	410.52
Q3	1578.48
IQR	1167.96
Lower Bound	-1341.41
Upper Bound	3330.41
Outliers Count	8
Eight TotalPrice outliers were identified using the IQR method. These represent genuine high-value customer transactions and were retained for business analysis.	

Outlier Summary

Variable	Outliers Detected
Quantity	0
UnitPrice	0
ItemsInCart	0
TotalPrice	8

The dataset remains suitable for analysis because the detected outliers are genuine business transactions rather than data entry errors.

Outlier Summary	
Analysis Type	Result
Quantity Outliers	0
UnitPrice Outliers	0
ItemsInCart Outliers	0
TotalPrice Outliers	8
Data Suitable for Analysis	Yes
Outliers Retained	Yes

Key Insights from Outlier Analysis

1. The IQR method successfully identified potential outliers.
2. No outliers were found in Quantity, UnitPrice, or ItemsInCart.
3. Eight outliers were detected in TotalPrice.
4. The detected outliers represent legitimate high-value transactions.
5. Revenue variation is influenced by a small number of high-value purchases.
6. The dataset remains reliable and suitable for business reporting and decision-making.

Business Insights
1. IQR method was used to identify potential outliers.
2. No outliers were found in Quantity, UnitPrice, or ItemsInCart.
3. Eight outliers were detected in TotalPrice.
4. These outliers represent genuine high-value transactions.
5. The dataset remains suitable for business analysis and reporting.
6. Revenue variation is influenced by a small number of legitimate high-value transactions.

Conclusion

The outlier analysis confirmed that the dataset is generally well-balanced and free from abnormal values in most variables. Only TotalPrice contained a small number of high-value outliers, which were retained because they represent genuine customer purchases. Therefore, the dataset is considered reliable for exploratory data analysis and business insight generation.

Final Conclusion

This Exploratory Data Analysis (EDA) project successfully examined the e-commerce dataset using Microsoft Excel, Power Query, Pivot Tables, and statistical analysis techniques.

The analysis provided valuable insights into customer purchasing behavior, product performance, payment preferences, referral effectiveness, and

revenue generation. Descriptive statistics helped summarize key numerical characteristics, while trend analysis identified important business patterns. Outlier detection using the IQR method revealed only a small number of high-value transactions in the TotalPrice variable, which were retained as valid business records. Overall, the dataset was found to be clean, reliable, and suitable for analytical reporting.

The findings generated through this project can support business decision-making, performance monitoring, and future data-driven strategies.